

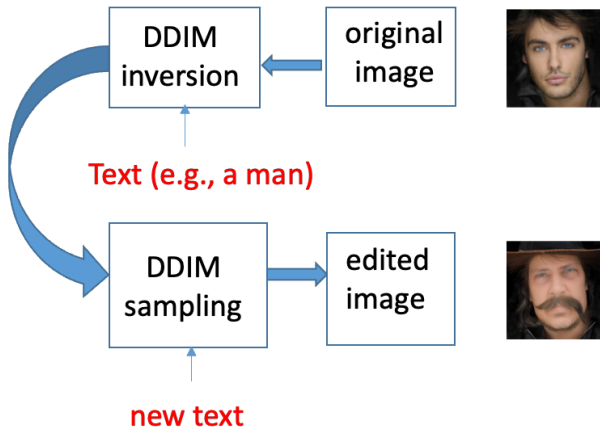
COMM114: Generative AI - Image Editing (part 2)

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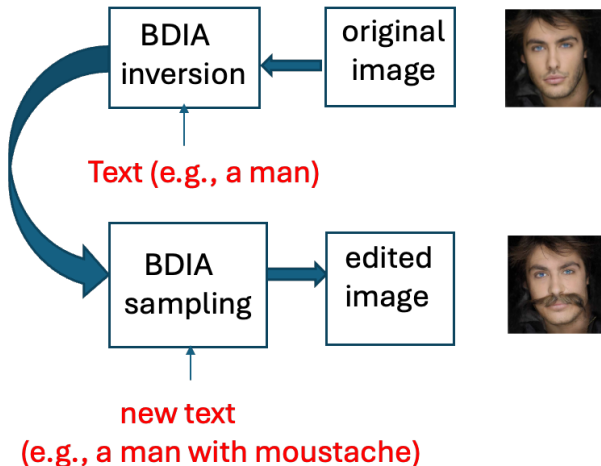
Recap from previous lecture

- DDIM-based Image Editing
 - DDIM inversion computes a latent representation of a clean image by solving the ODE from $t = 0$.
 - DDIM sampling computes a modified clean image with the obtained latent representation by solving the ODE until $t = 0$.



Recap from previous lecture (cont.)

- BDIA-based Image Editing
 - The BDIA technique attempts to reduce the approximation error introduced DDIM-based image editing.



Objective of this lecture

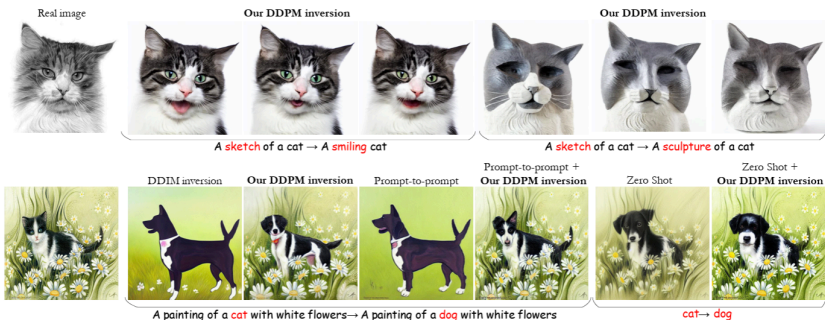
- Introduce DDPM based image editing
- Introduce Null-text based image editing

DDPM Based Image Editing (cont.)

An Edit Friendly DDPM Noise Space: Inversion and Manipulations

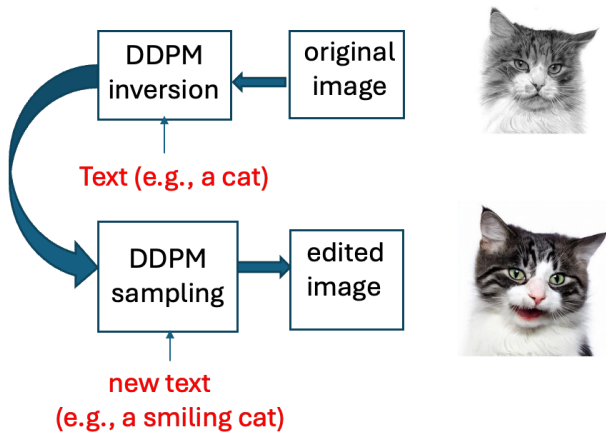
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0v3 [cs.CV] 9 Apr 2024



DDPM Based Image Editing (cont.)

- Pipeline



Background on Diffusion models

- A forward diffusion process (adding Gaussian noise to clean image)

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t} \mathbf{x} + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$$

- Reverse probability-flow ODE for sampling (generating an image by solving the ODE)

$$d\mathbf{z}_t = \overbrace{\left[\frac{d \log \sqrt{\bar{\alpha}_t}}{dt} \mathbf{z}_t + \left(\frac{1}{2\sqrt{1 - \bar{\alpha}_t}} \frac{d(1 - \bar{\alpha}_t)}{dt} - \frac{d \log \sqrt{\bar{\alpha}_t}}{dt} \sqrt{1 - \bar{\alpha}_t} \right) \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}, t}(\mathbf{z}_t) \right]}^{d(\mathbf{z}_t, t)} dt$$

where $\mathbf{z}_T \in \mathcal{N}(0, (1 - \bar{\alpha}_T) \mathbf{I})$

- $\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}, t}$ is the estimated Gaussian noise, which is the output a DNN model $\boldsymbol{\theta}$.

What is DDPM sampling?

- The DDPM sampling process starts from a random noise vector $\tilde{\mathbf{z}}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iteratively denoises it using the recursion.

$$\tilde{\mathbf{z}}_{t-1} = \hat{\boldsymbol{\mu}}_t(\tilde{\mathbf{z}}_t) + \sigma_t \boldsymbol{\epsilon}_t \quad t = T, \dots, 1,$$

where $\{\boldsymbol{\epsilon}_t\}_{t=1}^T$ are iid standard Gaussian noises, and

$$\hat{\boldsymbol{\mu}}_t(\tilde{\mathbf{z}}_t) = \sqrt{\bar{\alpha}_{t-1}} \frac{\tilde{\mathbf{z}}_t - \sqrt{1 - \bar{\alpha}} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}, t}(\tilde{\mathbf{z}}_t)}{\sqrt{\bar{\alpha}_t}} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}, t}(\tilde{\mathbf{z}}_t)$$

- The variable schedule for σ_t is taken to be

$$\sigma_t = \eta \beta_t (1 - \bar{\alpha}_{t-1}) / (1 - \bar{\alpha}_t), \quad \beta_t = 1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}$$

where $\eta \in [0, 1]$

- When $\eta = 0$, DDPM sampling reduces to DDIM sampling
- When $\eta = 1$, it corresponds to the original DDPM work.

What is DDPM sampling? (cont.)

- DDPM is a stochastic sampling method. At each iteration, some Gaussian noise is introduced in the denoising process to make the probability density function of the random vector $\tilde{z}_t$ smooth.
- When no Gaussian noise is introduced in the sampling process, it reduces to DDIM sampling.

DDPM Based Image Editing (cont.)

Algorithm 1 Procedure of DDPM-based image editing (cont.)

- 1: **Input:** real image x_0 ; ϕ : new text-prompt
 - 2: **for** $t = 1$ to T **do**
 - 3: $\tilde{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$
 - 4: $x_t \leftarrow \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\tilde{\epsilon}$ [diffusion forward process]
 - 5: **end for**
 - 6: Store $\{x_t\}_{t=1}^T$ in the memory
 - 7: **for** $t = T$ to 1 **do**
 - 8: $m_t = \tilde{\mu}_t(x_t, \phi)$ [next text prompt is used]
 - 9: $z_t \leftarrow (x_{t-1} - m_t)/\sigma_t$
 - 10: $x_{t-1} \leftarrow m_t + \sigma_t z_t$ [to avoid error accumulation]
 - 11: **end for**
 - 12: **Return:** z_1
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DDPM Based Image Editing (cont.)

- DDPM inversion essentially introduce Gaussian noises to the clean image x_0 .
- The introduced Gaussian noises are independent to each other.
- Some memory is needed to store the diffusion states $\{z_t\}_{t=1}^T$ obtained when we go from clean image to latent space.

DDPM Based Image Editing (cont.)



A photo of a horse in the mud



A photo of a zebra in the snow



A painting of a cat
with white flowers



A painting of a dog
with white flowers

DDPM Based Image Editing (cont.)

- project link for DDPM-based image editing
- huggingface demo

Null-text based image editing

Null-text Inversion for Editing Real Images using Guided Diffusion Models

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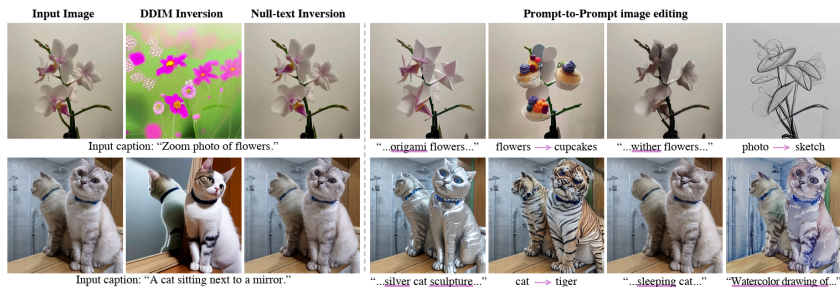


Figure 1. Null-text inversion for real image editing. Our method takes as input a real image (leftmost column) and an associated caption. The image is inverted with a DDIM diffusion model to yield a diffusion trajectory (second column to the left). Once inverted, we use the initial trajectory as a pivot for null-text optimization that accurately reconstructs the input image (third column to the left). Then, we can edit the inverted image by modifying only the input caption using the editing technique of Prompt-to-Prompt [16].

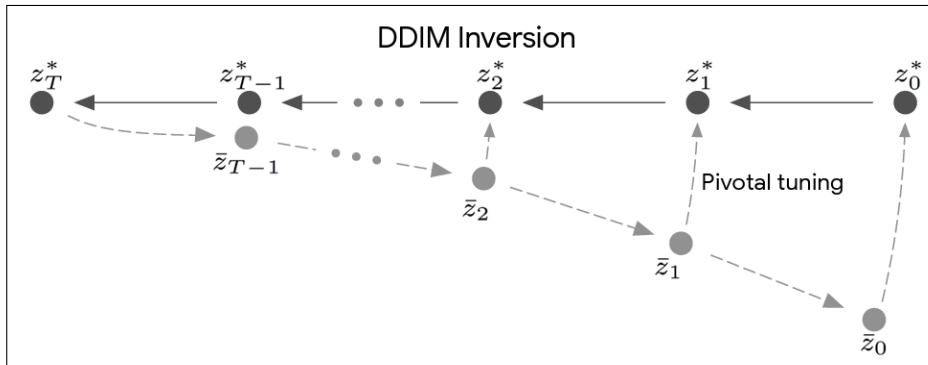
Null-text based image editing (cont.)

- The null-text-based image editing is an extension of DDIM-based image editing.
- Basic idea of null-text-based image editing
 1. Given a clean image x_0 , perform DDIM inversion with the original text-prompt to obtain a sequence of diffusion states $\{z_i^*\}_{t=0}^T$.
 2. Let $\bar{z}_T = z_T^*$. Then perform DDIM sampling with the original text-prompt to generate and optimize a sequence of diffusion states $\{\bar{z}\}_{i=1}^T$ such that each $\bar{z}_i$ is an accurate approximation of z_i^* .
 3. Then perform Prompt-to-Prompt technique to edit the image.

Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. "Prompt-to-prompt image editing with cross attention control". arXiv preprint arXiv:2208.01626, 2022.

Null-text based image editing (cont.)

- Suppose $\{z_i^*\}_{t=0}^T$ are obtained via DDIM inversion.
- Suppose $\{\bar{z}_i\}_{t=0}^T$ are obtained via DDIM sampling.
- Without **optimization**, the sampling sequence $\{\bar{z}_i\}_{t=0}^T$ might diverge from the sequence $\{z_i^*\}_{t=0}^T$ as t decreases from T to 0.



Null-text based image editing (cont.)

- At timestep t , we push z_{t-1} towards z_{t-1}^* by optimizing the following MSE over the unconditional embedding vector $\emptyset_t$.

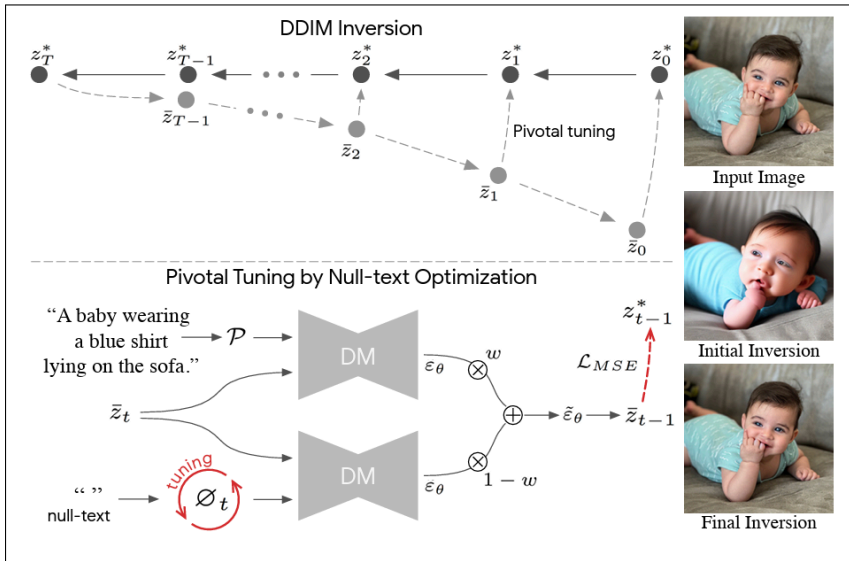
$$\emptyset_t^* = \arg \min_{\emptyset_t} \|z_{t-1}^* - z_{t-1}(\bar{z}_t, \emptyset_t, \mathcal{C})\|_2^2$$

- $\emptyset_t$: the unconditional embedding, which is treated as vector to be optimized.
- $\mathcal{C}$: the conditional embedding
- the optimization can be realized by gradient descent.
- After training, we let

$$\bar{z}_{t-1} = z_{t-1}(\bar{z}_t, \emptyset_t^*, \mathcal{C})$$

- The optimization of $\{\emptyset_t\}_{t=1}^T$ are performed sequentially starting from $t = T$ until $t = 0$.

Null-text based image editing (cont.)



Null-text based image editing (cont.)

- After obtaining $\{\emptyset_t^*\}_{t=T}^1$, one can then generate the edited image with $\{\emptyset_t^*\}_{t=T}^1$ and the new text embedding vector $\tilde{C}$ by running the DDIM sampling again:

$$\tilde{z}_T = z_T^*$$

$$\tilde{z}_{t-1} = z_{t-1}(\tilde{z}_t, \emptyset_t^*, \tilde{C}) \quad t = T, T-1, \dots, 1$$

- $\tilde{z}_0$ should be the edited image.

Null-text based image editing (cont.)



Null-text based image editing (cont.)

Input caption: “A baby wearing a blue shirt lying on the sofa.”



Input Image



“... blond baby...”



“... floral shirt...”



“... golden shirt...”



“... sleeping baby...”



“baby” → “robot”



“sofa” → “grass”



“sofa” → “ball pit”

References

- Null-text image editing paper
- DDPM-based image editing paper

Questions?